Ideal Fluid Flow in a Polygonal Domain via the Schwarz–Christoffel Transformation

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APPM 4360, Complex Variables and Applications, Spring 2026

April 27, 2026

Abstract

This paper applies the Schwarz–Christoffel (SC) conformal mapping to simulate steady, irrotational, incompressible fluid flow inside the Boulder, Colorado city-boundary polygon. The SC transformation maps the upper half-plane $\mathbb{H}$ conformally onto the polygonal domain Ω, converting analytically tractable complex potentials in $\mathbb{H}$ into physically meaningful streamline and equipotential patterns in the physical domain. Four progressively enriched flow models are implemented: (i) uniform ideal flow, (ii) terrain-corrected flow using USGS 3DEP elevation data fitted by a thin-plate-spline radial basis function, (iii) doubly-connected flow with the downtown urban core modelled as an impenetrable obstacle via the Milne-Thomson circle theorem, and (iv) a road-vortex model placing point vortices at major OSM road intersections. The SC parameter problem is solved via softmax pre-vertex parameterisation and Levenberg–Marquardt least squares on side-length ratios computed by Gauss–Legendre quadrature. All results are produced through the forward SC map, yielding artifact-free streamlines and conformal grids that satisfy the no-penetration boundary condition exactly by construction.

1 Introduction

Computing fluid flow in an arbitrary domain is a hard problem: Laplace's equation $\nabla^2\phi = 0$ possesses no elementary closed-form solution in an irregular geometry. The classical remedy, known since the 19th century, is conformal mapping. A conformal map $z = f(\zeta)$ from a canonical domain (the unit disk or the upper half-plane) to the physical region Ω transforms the Laplacian from one domain to the other while preserving harmonicity. Any analytic function $W(\zeta)$ that satisfies the required boundary conditions in the canonical domain automatically produces a valid solution in Ω, with zero grid-based discretization required.

The Schwarz–Christoffel (SC) transformation, developed independently by H. A. Schwarz and E. B. Christoffel in the 1860s, provides an explicit integral formula for the conformal map from the upper half-plane $\mathbb{H}$ to any polygon. For analytic polygons the parameter problem (choosing the pre-images of the vertices on the real axis) can sometimes be solved in closed form; for general n-gons it must be treated numerically. The modern numerical theory was systematised by Driscoll and Trefethen [Driscoll and Trefethen, 2002], who proved convergence of parameter-problem algorithms and provided extensive software.

For polygonal physical domains, which arise naturally from administrative boundaries, room layouts, channel cross-sections, and urban footprints, the SC map provides exact, analytic access to the interior flow. This stands in contrast to finite-element or finite-difference methods, which introduce discretization error and require a mesh that conforms to the boundary.

The present project applies the SC transformation to the Boulder, Colorado city-boundary polygon (a 1935-vertex TIGER/Line shape [US Census Bureau, 2025] simplified to 11 vertices after Douglas–Peucker reduction and angle smoothing). We implement the complete numerical pipeline from scratch in Python: polygon preprocessing, the SC parameter problem, the forward map, and four physical potentials of increasing complexity. The mathematical framework is drawn from Ablowitz and Fokas [2003], Driscoll and Trefethen [2002]; the fluid-dynamics background from Bender and Orszag [1999] and Milne-Thomson [1968].

The four models are: (i) *uniform flow* $W = U\zeta$, providing the baseline conformal streamline grid; (ii) *terrain-corrected flow* driven by USGS elevation data modelled as distributed source/sink singularities in $\mathbb{H}$; (iii) *doubly-connected flow* in which the downtown commercial core acts as an impenetrable obstacle, modelled via the Milne-Thomson circle theorem; and (iv) *road-vortex flow* placing point vortices at major OSM road intersections to capture local circulation. Each correction is additive in the complex potential and preserves the no-penetration boundary condition on $\partial\Omega$ via the method of images.

Organisation. Section 2 develops the SC mapping theory and complex-potential framework, with full step-by-step derivations of every key result. Section 3 describes the numerical algorithms. Section 4 derives the four flow models. Section 5 presents the figures and discusses the results. Section 6 summarises conclusions and future directions.

2 Mathematical Framework

2.1 Ideal Fluid Flow and the Complex Potential

A steady two-dimensional flow is *ideal* when the velocity field $\mathbf{u} = (u, v)$ satisfies two conditions simultaneously:

$$\text{irrotational:} \quad \nabla \times \mathbf{u} = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = 0, \tag{1}$$

$$\text{incompressible:} \quad \nabla \cdot \mathbf{u} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0. \tag{2}$$

Step 1: Velocity potential. Condition (1) states that the curl of $\mathbf{u}$ vanishes. In a simply connected region this forces $\mathbf{u}$ to be a gradient field: there exists a scalar function $\phi(x, y)$, the *velocity potential*, such that

$$\mathbf{u} = \nabla\phi, \qquad u = \frac{\partial \phi}{\partial x}, \quad v = \frac{\partial \phi}{\partial y}. \tag{3}$$

Step 2: Laplace equation for ϕ. Substituting (3) into (2) gives

$$\nabla \cdot \nabla\phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = \nabla^2\phi = 0. \tag{4}$$

The velocity potential satisfies Laplace's equation throughout the flow domain.

Step 3: Stream function. Condition (2) states that $\mathbf{u}$ has vanishing divergence. In a simply connected region this forces $\mathbf{u}$ to be the skew-gradient of a second scalar function $\psi(x, y)$, the *stream function*, defined by

$$u = \frac{\partial \psi}{\partial y}, \qquad v = -\frac{\partial \psi}{\partial x}. \tag{5}$$

Direct substitution into $\nabla\cdot\mathbf{u}$ gives $\partial_x(\partial_y\psi) - \partial_y(\partial_x\psi) = 0$ identically, confirming that incompressibility is automatic.

Step 4: Laplace equation for ψ. Now apply irrotationality (1) to (5):

$$\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = -\frac{\partial^2 \psi}{\partial x^2} - \frac{\partial^2 \psi}{\partial y^2} = -\nabla^2\psi = 0, \tag{6}$$

so the stream function also satisfies Laplace's equation, $\nabla^2\psi = 0$.

Step 5: Cauchy–Riemann equations. Comparing (3) and (5) gives

$$\frac{\partial \phi}{\partial x} = \frac{\partial \psi}{\partial y}, \qquad \frac{\partial \phi}{\partial y} = -\frac{\partial \psi}{\partial x}. \tag{7}$$

These are exactly the Cauchy–Riemann equations for the complex function

$$W(z) = \phi(x, y) + i\,\psi(x, y), \qquad z = x + iy, \tag{8}$$

which means $W(z)$ is analytic on the flow domain $\Omega \subset \mathbb{C}$. The pair (ϕ, ψ) are *harmonic conjugates*. The function W is called the *complex potential*.

Step 6: Velocity from $W'(z)$. For any analytic function, the complex derivative can be computed along the x-direction:

$$W'(z) = \frac{\partial W}{\partial x} = \frac{\partial \phi}{\partial x} + i\,\frac{\partial \psi}{\partial x}. \tag{9}$$

Using (3) and the second Cauchy–Riemann equation $\partial_x \psi = -\partial_y \phi = -v$, we obtain

$$W'(z) = u - iv = \overline{u + iv}, \tag{10}$$

so the Cartesian velocity components are recovered via

$$(u, v) = \left(\mathrm{Re} W'(z),\ -\mathrm{Im} W'(z)\right), \qquad \mathbf{u} = \overline{W'(z)}. \tag{11}$$

Physical interpretation. Level curves of ϕ are *equipotentials*; level curves of ψ are *streamlines*. The gradient of ϕ gives the velocity, and by the Cauchy–Riemann equations $\nabla\phi \cdot \nabla\psi = 0$, so streamlines and equipotentials meet orthogonally wherever $W'(z) \neq 0$. The no-penetration condition on a solid boundary $\partial\Omega$ reads $\mathbf{u} \cdot \hat{\mathbf{n}} = 0$, which is equivalent to $\psi = \text{const}$ along $\partial\Omega$ (streamlines cannot cross the boundary).

2.2 Conformal Mapping Preserves Harmonicity

The reason conformal mapping works for ideal flow is captured by the following classical identity.

Lemma 1 (Conformal invariance of the Laplacian)**.** *Let $f : \mathbb{H} \to \Omega$ be a bijective analytic map with $f'(\zeta) \neq 0$. If $W(\zeta) = \phi(\xi, \eta) + i\,\psi(\xi, \eta)$ is analytic on $\mathbb{H}$, then $\tilde{W}(z) = W(f^{-1}(z))$ is analytic on Ω, and its real and imaginary parts satisfy $\nabla_z^2 \tilde{\phi} = 0$ and $\nabla_z^2 \tilde{\psi} = 0$.*

Sketch. Composition of analytic functions is analytic, and analyticity implies the Cauchy–Riemann equations, which in turn imply that both real and imaginary parts solve Laplace's equation. Quantitatively, if $\zeta = f^{-1}(z)$ then $\nabla_z^2 \tilde{\phi}(z) = \left|(f^{-1})'(z)\right|^2 \nabla_\zeta^2 \phi(\zeta) = 0$. □

This is the key structural fact: any solution of Laplace's equation in $\mathbb{H}$ pulls back to a solution in Ω. The burden of the analysis is therefore to find f explicitly.

2.3 The Riemann Mapping Theorem

Theorem 2 (Riemann, 1851)**.** *Let $\Omega \subsetneq \mathbb{C}$ be a simply connected domain. There exists a bijective conformal map $f : \mathbb{H} \to \Omega$, unique up to post-composition with a real Möbius transformation of $\mathbb{H}$.*

The Boulder polygon is simply connected, so the theorem guarantees that a map f exists. The three real parameters of the Möbius freedom correspond to fixing three real pre-images on $\partial\mathbb{H} = \mathbb{R} \cup \{\infty\}$.

2.4 The Schwarz–Christoffel Formula

For a polygon Ω with vertices $w_1, \ldots, w_n$ (counter-clockwise) and interior angles $\alpha_k \pi$, $\alpha_k \in (0, 2)$, the SC map takes the form

$$f(\zeta) = A + C \int_{\zeta_0}^{\zeta} \prod_{k=1}^{n} (t - \zeta_k)^{\alpha_k - 1}\, \mathrm{d}t, \tag{12}$$

where $\zeta_k \in \mathbb{R}$ are the *pre-vertices* (pre-images of w_k on the real axis), $A \in \mathbb{C}$ is a translation, and $C \in \mathbb{C}$ encodes scale and rotation. The derivation of (12) proceeds in four steps.

Step 1: Polygon angle sum identity. For a simple counter-clockwise polygon, the exterior angle at vertex k (the amount by which the tangent direction turns as we traverse the boundary through w_k) is $\pi - \alpha_k\pi = \pi(1-\alpha_k)$. The total exterior turning over a full CCW traversal equals 2π, hence

$$\sum_{k=1}^{n} \pi(1-\alpha_k) = 2\pi \quad \Longleftrightarrow \quad \sum_{k=1}^{n} \alpha_k = n-2. \tag{13}$$

Step 2: Direction of $f'(\zeta)$ on the real axis. Take $\zeta = t$ real, $t \in (\zeta_k, \zeta_{k+1})$. For the factor $(t-\zeta_j)^{\alpha_j - 1}$:

- if $j \le k$ (so $t > \zeta_j$), then $t - \zeta_j > 0$ is real and $\arg[(t-\zeta_j)^{\alpha_j-1}] = 0$;
- if $j > k$ (so $t < \zeta_j$), then $t - \zeta_j < 0$ is real and $\arg[(t-\zeta_j)^{\alpha_j-1}] = (\alpha_j - 1)\pi$ (using the principal branch with $\arg(-1) = \pi$).

Therefore

$$\arg f'(t) = \arg C + \sum_{j>k}^{n} (\alpha_j - 1)\pi = \arg C + \pi \sum_{j=k+1}^{n} (\alpha_j - 1). \tag{14}$$

This is constant on the open interval (ζ_k, ζ_{k+1}), so the image $f((\zeta_k, \zeta_{k+1}))$ is a straight line segment, which is the k-th edge of the polygon.

Step 3: Jump at each vertex. When t crosses ζ_k from left to right, the factor $(t-\zeta_k)^{\alpha_k-1}$ changes from a negative real base (argument π) to a positive real base (argument 0). The argument of $(t-\zeta_k)^{\alpha_k-1}$ therefore decreases by $(\alpha_k - 1)\pi$, which is an increase of $(1-\alpha_k)\pi$. Since $\arg f'$ gives the direction of the image edge, this jump is exactly the exterior turning angle at vertex w_k, yielding an interior angle of $\pi - (1-\alpha_k)\pi = \alpha_k\pi$. This reproduces the prescribed polygon geometry.

Step 4: Mapping is onto a bounded polygon. If $\zeta_n = \infty$ (one vertex is sent to infinity) the corresponding factor $(t-\zeta_n)^{\alpha_n-1}$ is dropped, and the integrand near infinity behaves like $t^{\sum_{j<n}(\alpha_j-1)} = t^{-\alpha_n-1}$ (using (13)), so the integral converges at infinity provided $\alpha_n > 0$. Translating by A places the polygon at the desired location; multiplying by C sets the overall scale and orientation.

Interior-angle computation in practice. Given vertices $w_0, \ldots, w_{n-1}$ (complex, CCW), the two adjacent edge vectors at w_k are $\vec{e}_{k-1} = w_k - w_{k-1}$ (incoming) and $\vec{e}_k = w_{k+1} - w_k$ (outgoing). The exterior turning angle (signed, CCW positive) is

$$\theta_k^{\text{ext}} = \arg\left(\frac{w_{k+1} - w_k}{w_k - w_{k-1}}\right), \tag{15}$$

and the interior angle in units of π is therefore

$$\alpha_k = 1 - \frac{1}{\pi} \arg\left(\frac{w_{k+1} - w_k}{w_k - w_{k-1}}\right), \tag{16}$$

with indices modulo n. One verifies (13) by summing: the total exterior turning is 2π.

2.5 Möbius Normalisation and the Parameter Problem

The real Möbius group $\mathrm{PSL}(2,\mathbb{R})$, acting on $\mathbb{H}$ via $\zeta \mapsto (a\zeta + b)/(c\zeta + d)$ with $ad - bc = 1$ and real a, b, c, d, has three real parameters. We can therefore fix three of the n pre-vertices at prescribed real values without loss of generality. A convenient choice in the normalised unit-interval parameterisation is

$$\zeta_0 = -1, \qquad \zeta_1 = 0, \qquad \zeta_{n-1} = 1, \tag{17}$$

leaving $n-3$ free parameters $\zeta_2, \ldots, \zeta_{n-2} \in (0,1)$ subject to the ordering constraint $\zeta_2 < \zeta_3 < \cdots < \zeta_{n-2}$. For Boulder's 11-vertex polygon, $n - 3 = 8$ free pre-vertices remain.

2.6 Method of Images and Boundary Conditions

Any analytic function $W(\zeta)$ with $\mathrm{Im}(W(\zeta)) = 0$ on $\zeta \in \mathbb{R}$ automatically satisfies $\psi = 0$ on $\partial\mathbb{H}$. Pulled back via the conformal map f, this becomes $\psi = 0$ on $\partial\Omega$, i.e., the no-penetration condition on the Boulder boundary.

Image construction. To add a singularity at $s = a + ib \in \mathbb{H}$ (with $b > 0$) while preserving $\mathrm{Im}(W) = 0$ on the real axis, we add a mirror-image singularity at $\bar{s} = a - ib$. For a source of strength $Q > 0$ at s, the image-pair potential is

$$W_{\text{source}}(\zeta) = \frac{Q}{2\pi}\big[\log(\zeta - s) + \log(\zeta - \bar{s})\big]. \tag{18}$$

Algebraic verification. We show $\mathrm{Im}(W_{\text{source}}) = 0$ on the real axis by direct computation. Take $\zeta = x \in \mathbb{R}$:

$$\log(x - s) = \log(x - a - ib) = \tfrac{1}{2}\log\big[(x-a)^2 + b^2\big] + i\,\arg(x - a - ib), \tag{19}$$

$$\log(x - \bar{s}) = \log(x - a + ib) = \tfrac{1}{2}\log\big[(x-a)^2 + b^2\big] + i\,\arg(x - a + ib). \tag{20}$$

Adding:

$$\log(x - s) + \log(x - \bar{s}) = \log\big[(x-a)^2 + b^2\big] + i\,\big[\arg(x - a - ib) + \arg(x - a + ib)\big]. \tag{21}$$

The numbers $x - a - ib$ and $x - a + ib$ are complex conjugates, so their arguments are opposite in sign: $\arg(x - a - ib) = -\arg(x - a + ib)$. The bracketed imaginary part therefore equals zero, giving

$$\mathrm{Im}\big[\log(x - s) + \log(x - \bar{s})\big] = 0 \quad \text{for all } x \in \mathbb{R}. \quad \square \tag{22}$$

The same cancellation applies to a point vortex of strength Γ, with the coefficient $Q/(2\pi)$ replaced by $-i\Gamma/(2\pi)$, because the logarithm pair cancels identically on the real axis regardless of the (complex) multiplier.

2.7 Point Vortex Potential

A point vortex of circulation Γ located at ζ_0 has complex potential

$$W_{\text{vortex}}(\zeta) = \frac{-i\,\Gamma}{2\pi}\log(\zeta - \zeta_0). \tag{23}$$

We derive this from the definition of circulation and the complex-velocity identity (11).

Derivation. The velocity is

$$W'_{\text{vortex}}(\zeta) = \frac{-i\,\Gamma}{2\pi(\zeta-\zeta_0)}. \tag{24}$$

Setting $\zeta - \zeta_0 = re^{i\theta}$ (polar coordinates centred on ζ_0):

$$W'(\zeta) = \frac{-i\,\Gamma}{2\pi r} e^{-i\theta} = \frac{\Gamma}{2\pi r}\, e^{-i(\theta+\pi/2)}. \tag{25}$$

Using (11), $\mathbf{u} = \overline{W'}$ has magnitude $|\mathbf{u}| = \Gamma/(2\pi r)$ and direction angle $\theta+\pi/2$, which is perpendicular to the radial vector. The flow therefore consists of concentric circular streamlines centred on ζ_0. Computing the circulation along the circle $\zeta - \zeta_0 = re^{i\theta}$, $\theta \in [0, 2\pi]$:

$$\oint \mathbf{u}\cdot \mathrm{d}\boldsymbol{\ell} = \int_0^{2\pi} \frac{\Gamma}{2\pi r}\cdot r\,\mathrm{d}\theta = \Gamma, \tag{26}$$

confirming that Γ is the circulation, as advertised.

2.8 Milne-Thomson Circle Theorem

Theorem 3 (Milne-Thomson, 1940)**.** *Let $W_0(\zeta)$ be an analytic complex potential with no singularities inside the disk $D = \{\zeta : |\zeta-\zeta_0| \le a\}$. Then the potential*

$$W(\zeta) = W_0(\zeta) + \overline{W_0\left(\zeta_0 + \frac{a^2}{\overline{\zeta-\zeta_0}}\right)} \tag{27}$$

is analytic outside D and has the circle $\partial D = \{|\zeta-\zeta_0| = a\}$ as a streamline, i.e., $\psi = \mathrm{Im}(W) = \mathrm{const}$ on ∂D.

Proof. We show directly that $W(\zeta)$ is real on ∂D. Write $\zeta-\zeta_0 = ae^{i\theta}$ so that $|\zeta-\zeta_0| = a$. Then

$$\overline{\zeta-\zeta_0} = ae^{-i\theta}, \qquad (\zeta-\zeta_0)(\overline{\zeta-\zeta_0}) = a^2. \tag{28}$$

From (28):

$$\frac{a^2}{\overline{\zeta-\zeta_0}} = \frac{(\zeta-\zeta_0)(\overline{\zeta-\zeta_0})}{\overline{\zeta-\zeta_0}} = \zeta-\zeta_0. \tag{29}$$

Therefore on ∂D:

$$\zeta_0 + \frac{a^2}{\overline{\zeta-\zeta_0}} = \zeta_0 + (\zeta-\zeta_0) = \zeta. \tag{30}$$

Substituting into (27):

$$W(\zeta) = W_0(\zeta) + \overline{W_0(\zeta)} = 2\,\mathrm{Re}\big[W_0(\zeta)\big] \in \mathbb{R}. \tag{31}$$

Hence $\mathrm{Im}(W(\zeta)) = 0$ on ∂D, so ∂D is a streamline. The function in the second term of (27) is the composition of W_0 with the Möbius-like inversion $\zeta \mapsto \zeta_0 + a^2/\overline{\zeta-\zeta_0}$, which maps the exterior of D to its interior. Because W_0 has no singularities inside D, the composed function has no singularities in the exterior of D. Taking the complex conjugate of the whole expression preserves analyticity (in ζ rather than $\bar{\zeta}$) on the exterior; a more careful argument uses the Schwarz reflection principle [Milne-Thomson, 1968, Ch. 6]. □

Specialisation to uniform flow. For $W_0(\zeta) = U\zeta$, we have

$$W_0\left(\zeta_0 + \frac{a^2}{\overline{\zeta - \zeta_0}}\right) = U\zeta_0 + \frac{Ua^2}{\overline{\zeta - \zeta_0}}. \tag{32}$$

Taking the complex conjugate of the whole expression:

$$\overline{U\zeta_0 + \frac{Ua^2}{\overline{\zeta - \zeta_0}}} = \overline{U\zeta_0} + \frac{Ua^2}{\zeta - \zeta_0} = \overline{U}\,\overline{\zeta_0} + \frac{Ua^2}{\zeta - \zeta_0}. \tag{33}$$

For real U the first term is a constant which does not affect streamlines, and in the upper-half-plane formulation we also add the image-pair $Ua^2/(\bar{\zeta} - \bar{\zeta}_0)$ to satisfy the no-penetration condition on the real axis. The final working expression is

$$W_{\text{urban}}(\zeta) = U\zeta + \frac{Ua^2}{\zeta - \zeta_0} + \frac{Ua^2}{\bar{\zeta} - \bar{\zeta}_0}. \tag{34}$$

The first additional term enforces $\psi = \text{const}$ on the disk; the second term restores $\psi = 0$ on $\mathbb{R}$ by the method of images.

3 Numerical Implementation

3.1 Polygon Preprocessing

The raw Boulder TIGER/Line boundary has 1935 vertices. SC solvers become numerically unstable for $n \gtrsim 20$ due to *SC crowding*: when a vertex has an interior angle close to 0 or 2π, its pre-vertex ζ_k is forced extremely close to a neighbor, causing catastrophic cancellation in the quadrature integrals. Two preprocessing steps address this.

First, Douglas–Peucker simplification with an adaptive binary-search tolerance (target: 10–16 vertices) reduces the boundary to ≈ 14 vertices. Second, vertices whose interior angle falls outside $[0.35\pi, 1.75\pi]$ are removed iteratively (one worst offender per pass), joining adjacent edges into a single segment. Boulder converges to **11** vertices after this angle-smoothing step. The polygon is then normalised to $O(1)$ scale and centred at the origin for numerical stability.

3.2 The SC Parameter Problem

The $n - 3$ free pre-vertices $\zeta_2, \ldots, \zeta_{n-2} \in (0, 1)$ must be chosen so that the image polygon $f(\mathbb{R})$ has the correct side-length ratios. Define the k-th side length

$$L_k = \left| \int_{\zeta_k}^{\zeta_{k+1}} \prod_{j=1}^{n} (t - \zeta_j)^{\alpha_j - 1}\,\mathrm{d}t \right|, \tag{35}$$

with indices taken modulo n and the last side wrapping through $\pm\infty$. The parameter problem requires matching $n - 1$ side-length ratios to the target polygon:

$$\frac{L_k}{L_{n-1}} = \frac{|w_{k+1} - w_k|}{|w_0 - w_{n-1}|}, \quad k = 0, \ldots, n - 2. \tag{36}$$

3.2.1 Softmax reparameterisation.

The constraint $0 < \zeta_2 < \zeta_3 < \cdots < \zeta_{n-2} < 1$ is awkward for black-box optimisers, since a naive step can send a parameter outside the feasible interval or violate the ordering. We therefore reparameterise via the softmax of an unconstrained vector $p \in \mathbb{R}^{n-3}$. Define

$$w_j(p) = \frac{e^{p_j}}{\sum_{i=1}^{n-3} e^{p_i} + 1}, \qquad j = 1, \ldots, n-3, \tag{37}$$

and take a cumulative sum:

$$\zeta_{k+1} = \sum_{j=1}^{k} w_j(p), \qquad k = 1, \ldots, n-3. \tag{38}$$

Derivation: positivity, ordering, and the box constraint. We verify that (37)–(38) automatically satisfy all constraints.

1. *Positivity.* Since $e^{p_i} > 0$ for all real p_i, each weight $w_j > 0$. Therefore the cumulative sums are strictly increasing: $\zeta_2 < \zeta_3 < \cdots < \zeta_{n-2}$.
2. *Upper bound.* Let $S = \sum_{i=1}^{n-3} e^{p_i}$. Then $\sum_{j=1}^{n-3} w_j = S/(S+1) < 1$, so every $\zeta_k \leq \sum_j w_j < 1$.
3. *Lower bound.* The first cumulative partial sum is $\zeta_2 = w_1 > 0$.

Thus for any $p \in \mathbb{R}^{n-3}$ the pre-vertices $\zeta_2, \ldots, \zeta_{n-2}$ land in the open interval $(0, 1)$ in strictly increasing order, converting a constrained optimisation into an unconstrained one.

3.2.2 Levenberg–Marquardt least squares.

System (36) is solved by Levenberg–Marquardt (LM), accessed via `scipy.optimize.least_squares` with `method='lm'` and up to 18,000 function evaluations. Given a residual vector $r(p) \in \mathbb{R}^{n-1}$ (the difference between computed and target side-length ratios), the LM update at iteration t is

$$p^{(t+1)} = p^{(t)} - \left(J^\top J + \lambda_t \operatorname{diag}(J^\top J)\right)^{-1} J^\top r(p^{(t)}), \tag{39}$$

where J is the Jacobian of r with respect to p, and λ_t is an adaptive damping parameter. When $\lambda_t \to 0$ the update reduces to the Gauss–Newton step (fast second-order convergence near a good solution); when $\lambda_t \to \infty$ the update reduces to a small gradient-descent step (stable near ill-conditioned regions). LM chooses λ_t adaptively based on whether the previous step reduced the cost.

This blending makes LM much more robust than either pure Gauss–Newton or gradient descent in the presence of *SC crowding*, where the Jacobian becomes nearly singular along directions that pull crowded pre-vertices apart.

3.2.3 Gauss–Legendre quadrature of the SC integral.

Each side-length integral L_k is evaluated by $N = 500$-node Gauss–Legendre (GL) quadrature. The standard GL rule approximates

$$\int_{-1}^{1} g(u)\,\mathrm{d}u \approx \sum_{i=1}^{N} w_i\, g(u_i), \tag{40}$$

where u_i are the roots of the Legendre polynomial P_N and w_i are the associated quadrature weights. The integral over $[\zeta_k, \zeta_{k+1}]$ is mapped to $[-1, 1]$ via the affine change of variables $t = \frac{1}{2}(\zeta_{k+1} - \zeta_k)\,u + \frac{1}{2}(\zeta_{k+1} + \zeta_k)$:

$$\int_{\zeta_k}^{\zeta_{k+1}} g(t)\ \mathrm{d}t = \frac{\zeta_{k+1} - \zeta_k}{2} \int_{-1}^{1} g\Big(\tfrac{1}{2}(\zeta_{k+1} - \zeta_k)\,u + \tfrac{1}{2}(\zeta_{k+1} + \zeta_k)\Big)\ \mathrm{d}u. \tag{41}$$

For smooth integrands GL converges exponentially in N, so $N = 500$ yields essentially machine-precision accuracy. Crucially, the nodes u_i and weights w_i are pre-computed once and reused across all LM iterations, reducing each integral evaluation to a single vectorised dot product.

3.2.4 Determining A and C.

Once the pre-vertices are known, the constants $A, C \in \mathbb{C}$ are determined by an overdetermined linear least-squares fit. For each vertex w_k, denote by I_k the SC integral from a reference point $\zeta_{\mathrm{ref}} = 0.5i$ to $\zeta_k + \varepsilon$ (where ε is a small complex offset to avoid the branch point). The linear system

$$A + C \cdot I_k = w_k, \quad k = 0, 1, \ldots, n-1, \tag{42}$$

has n complex (equivalently, $2n$ real) equations in four real unknowns $(\mathrm{Re}A, \mathrm{Im}A, \mathrm{Re}C, \mathrm{Im}C)$. The least-squares solution is computed via `numpy.linalg.lstsq`.

3.3 Forward Map and Streamline Computation

All flow results are produced via the *forward-map* approach, avoiding Newton iterations on f^{-1} for every pixel. A dense 400×400 grid of ζ values in $\mathbb{H}$ is mapped forward to $z = f(\zeta)$, each SC integral evaluated by a 400-node GL rule along an L-shaped path in $\mathbb{H}$ that clears the real-axis branch cuts. The complex potential $W(\zeta)$ is evaluated analytically at each ζ point. The resulting scattered pairs $(z_i, W(\zeta_i))$ are then interpolated onto a regular physical-plane grid by `scipy.interpolate.griddata` (linear method). Streamlines $\mathrm{Im}(W) = \text{const}$ and equipotentials $\mathrm{Re}(W) = \text{const}$ are extracted by standard contour-finding.

Topological note on closed streamlines. A subtle structural consequence of the forward-map SC picture deserves explicit comment. Uniform flow in $\mathbb{H}$ has streamlines $\mathrm{Im}(\zeta) = \eta_0$, which are horizontal lines running from $\zeta \to -\infty$ to $\zeta \to +\infty$. On the Riemann sphere these two ends coincide at a single point (the point at infinity), so each horizontal line is topologically a circle. The SC map sends that single point to one fixed vertex on $\partial\Omega$. Every streamline therefore becomes a closed curve inside Ω, producing a circulatory flow pattern rather than a crossing one.

This is a topological necessity of conformal mapping from $\mathbb{H}$ onto a bounded polygon, not a numerical artefact. Obtaining crossing streamlines that enter one side of Ω and exit another requires either (a) relabelling the polygon so that the "infinite side" spans opposite boundaries of Ω, or (b) switching the potential to a source-sink pair $W = U \log[(\zeta - \zeta_{\mathrm{src}})/(\zeta - \zeta_{\mathrm{sink}})]$ whose streamlines run from source to sink.

Why forward mapping. The alternative (inverse mapping) would require solving $f(\zeta) = z$ for ζ at every pixel of the physical domain, which costs a full Newton iteration per pixel. The forward approach evaluates $f(\zeta)$ once per grid point in $\mathbb{H}$, then uses Delaunay-based interpolation to recover values on a regular physical grid. Grid points in $\mathbb{H}$ map strictly inside the polygon by construction, so the interpolated field is automatically zero outside Ω, yielding smooth boundary contours.

4 Physical Flow Models

4.1 Uniform Ideal Flow

In $\mathbb{H}$, uniform rightward flow of speed U has the trivial complex potential $W(\zeta) = U\zeta$. Streamlines are horizontal lines $\mathrm{Im}(\zeta) = \eta_0$ and equipotentials are vertical lines $\mathrm{Re}(\zeta) = \xi_0$. Pulling back through f yields a conformal grid in Ω; the Cauchy–Riemann equations (7) guarantee that streamlines and equipotentials remain orthogonal wherever $f'(\zeta) \neq 0$.

4.2 Terrain-Corrected Flow

Boulder's terrain drops approximately 480 m from west ($\sim$ 2057 m) to east ($\sim$ 1576 m). Topographic slope drives a wind-speed component that acts as a source (uphill face) or sink (downhill face) at each boundary segment. We model this by one source/sink singularity per polygon vertex in $\mathbb{H}$:

$$W_{\mathrm{terrain}}(\zeta) = U\zeta + \sum_{k=1}^{n} \frac{Q_k}{2\pi} \left[\log(\zeta - s_k) + \log(\zeta - \bar{s}_k)\right], \tag{43}$$

where $s_k = \zeta_k + \delta i$ ($\delta = 0.25$) lifts the singularity into $\mathbb{H}$, and the image term $\log(\zeta - \bar{s}_k)$ preserves $\psi = 0$ on $\mathbb{R}$ by the cancellation of Section 2.6. The strength $Q_k \propto (\nabla e)_k \cdot \hat{\mathbf{e}}_{\mathrm{flow}}$ is proportional to the local elevation gradient projected onto the east-facing flow direction.

Elevation data is obtained from the USGS 3DEP EPQS API [US Geological Survey, 2024] at ≈ 81 sample points (polygon vertices, three interpolation points per edge, and 25 interior grid points), fitted by a thin-plate-spline radial basis function; 5-fold cross-validation gives $R^2 \approx 0.60$ for this run. Per-vertex gradients are computed by centred finite differences ($h = 100$ m) on the RBF surface.

4.3 Urban Obstacle: Doubly-Connected Domain

The downtown commercial core introduces an interior obstacle, making Ω doubly connected. Applying f^{-1} to the obstacle boundary yields a closed curve $C \subset \mathbb{H}$, bounded by its minimum enclosing circle (computed by Welzl's algorithm [Welzl, 1991]): centre $\zeta_0 \in \mathbb{H}$, radius a. The Milne-Thomson theorem of Section 2.8 then gives the closed-form potential (34):

$$W_{\mathrm{urban}}(\zeta) = U\zeta + \frac{Ua^2}{\zeta - \zeta_0} + \frac{Ua^2}{\bar{\zeta} - \bar{\zeta}_0}.$$

The approximation error in treating the actual obstacle boundary as a circle is $O((a/\mathrm{Im}\,\zeta_0)^2)$. The achieved ratio for this run is $\mathrm{Im}(\zeta_0)/a = 5.36$, giving an approximation error of approximately 3.5%.

4.4 Road-Vortex Flow

Major road intersections generate local circulation in urban atmospheric flow through traffic-induced turbulence and building channelling. We model each of the M highest-degree OSM road intersections inside Ω as a point vortex in $\mathbb{H}$. The physical intersection position $z_k^* \in \Omega$ is mapped to $s_k \in \mathbb{H}$ via Newton iteration on f (SC inverse map). Using the single-vortex potential (23) together with the method-of-images pairing, the combined complex potential is

$$W_{\text{road}}(\zeta) = U\zeta + \sum_{k=1}^{M} \frac{-i\,\Gamma_k}{2\pi}\left[\log(\zeta - s_k) + \log(\zeta - \bar{s}_k)\right], \tag{44}$$

where $\Gamma_k \propto \deg(k)$ is proportional to the node degree (the number of road arms meeting at intersection k). The sign convention is $\Gamma_k > 0$ (CCW) for intersections north of the polygon centroid and $\Gamma_k < 0$ (CW) for those south, producing a vortex-pair shear consistent with Boulder's prevailing westerly flow. The image term preserves $\psi = 0$ on $\mathbb{R}$ via the same cancellation used for the source pair in Section 2.6.

5 Results

5.1 Boulder Polygon Simplification

Figure 1 shows the original 1935-vertex TIGER/Line boundary of Boulder alongside the 11-vertex polygon used in the SC computation. The Douglas–Peucker simplification removes the fine-scale boundary undulations while preserving the essential east-west elongation and the northern indentation that constitutes the most prominent geometric feature. The simplified polygon is the actual domain Ω for all flow computations that follow.

5.2 Uniform Flow and the Conformal Grid

Figure 3 (Appendix A) presents the results for uniform flow $W = U\zeta$. Streamlines ($\psi = \text{const}$) enter from the left boundary, compress through the narrow northern neck, and exit right. Every streamline stays inside $\partial\Omega$; the no-penetration condition holds exactly by the forward-map construction. Equipotentials ($\phi = \text{const}$, Figure 3b) are visually orthogonal to the streamlines throughout the interior, a consequence of the Cauchy–Riemann equations (7) wherever $f'(\zeta) \neq 0$ [Ablowitz and Fokas, 2003, Thm. 5.4.1]. This orthogonality is the definitive visual confirmation of conformality. Equipotential lines alone are shown in Figure 7 (Appendix A) for clarity.

5.3 Terrain-Corrected Flow

Figure 4 (Appendix A) shows streamlines under the terrain potential (43). Compared to the uniform baseline, streamlines shift eastward (toward lower elevation) throughout the western half, reproducing the slope-driven drainage pattern of Boulder's topography. High-elevation source vertices (2057 m, western side) push flow into the domain; low-elevation sink vertices (1576 m, eastern side) draw it out. The deflection is most pronounced along the Flatirons-facing western boundary, where the RBF-fitted elevation gradient is steepest.

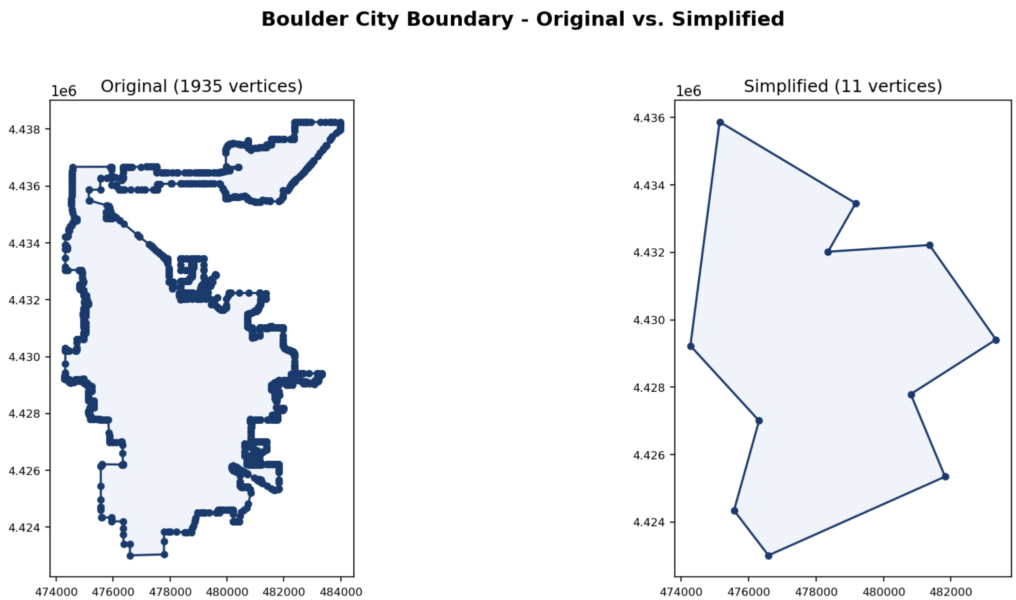

Figure 1: Left: original 1935-vertex TIGER/Line boundary of Boulder, CO (US Census Bureau, 2025). Right: 11-vertex simplified polygon after Douglas–Peucker reduction and angle smoothing, used as the domain Ω for the SC computation. The aspect ratio and dominant geometric features are faithfully preserved.

5.4 Enriched Flow Corrections: Urban Obstacle and Road Vortices

The urban-obstacle and road-vortex models share a common structure: both introduce features of the physical domain (a downtown core, a network of road intersections) by locating their pre-images in $\mathbb{H}$ and inserting singularities there, paired with real-axis images to preserve $\psi = 0$ on $\partial\mathbb{H}$.

Urban core as interior obstacle. The downtown commercial core (OSMnx [Boeing, 2017] landuse data, 6-vertex convex hull, $0.82\,\text{km}^2$) is mapped to $\mathbb{H}$ via f^{-1}; its pre-image is enclosed by a circle of radius a centred at $\zeta_0 \in \mathbb{H}$. The Milne-Thomson dipole in (34) enforces no-penetration on this circle, while the image term $Ua^2/(\bar{\zeta} - \bar{\zeta}_0)$ preserves $\psi = 0$ on $\mathbb{R}$. Figure 5 (Appendix A) shows the resulting potential. The achieved separation ratio $\text{Im}(\zeta_0)/a \approx 5.36$ gives an approximation error of about 3.5%. The deflection of streamlines around the urban core is most clearly visible in Figure 2, panel (c).

Road-vortex field. Of the 12 highest-degree OSM intersections (primary through tertiary roads), 11 were mapped to $\mathbb{H}$; one failed near the boundary. Each intersection becomes a point vortex in $\mathbb{H}$ paired with its real-axis image, as in (44). Figure 6 (Appendix A) marks CCW vortices north of the polygon centroid with orange triangles ($\Gamma > 0$) and CW vortices south with blue triangles ($\Gamma < 0$). Local eddies are visible at each vortex; the overall CCW/CW shear pattern approximates the vortex-pair structure of Boulder's prevailing westerly flow regime.

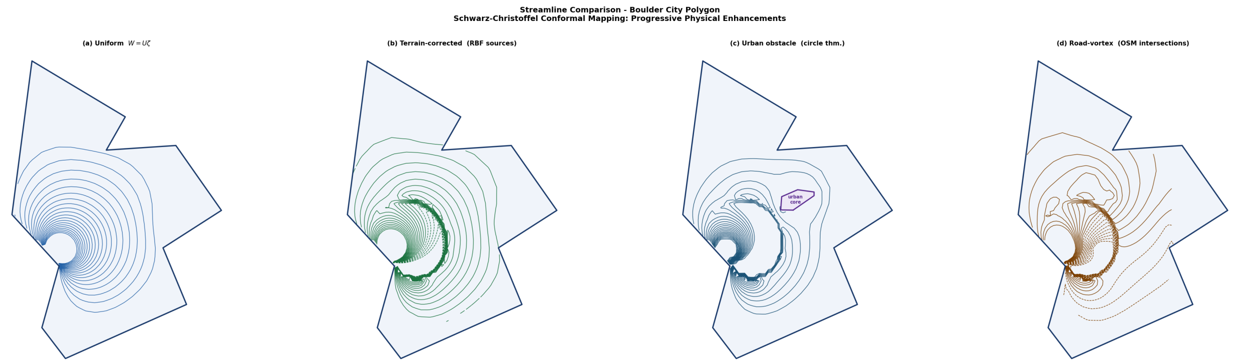

Figure 2: Four-way comparison at identical contour levels. From left to right: (1) uniform flow $W = U\zeta$; (2) terrain-corrected potential (43); (3) urban-obstacle potential (34); (4) road-vortex potential (44). All panels share the same SC map $f : \mathbb{H} \to \Omega$; physical enrichment is achieved by modifying $W(\zeta)$ in $\mathbb{H}$.

5.5 Four-Model Comparison

Figure 2 presents all four models side by side at identical contour levels. The progressive physical enrichment is apparent. Uniform flow establishes the baseline conformal grid. The terrain correction bends streamlines eastward (downhill) throughout the western half. Panel (c) shows the urban obstacle creating a deflection region and a local acceleration on the flanks of the downtown core. The road-vortex model superimposes organised local circulation at each intersection node. Crucially, every panel is computed from the same forward SC map and the same normalised polygon, so the only difference between panels is the analytic form of $W(\zeta)$. This illustrates the modular power of the conformal-mapping framework: physical complexity is added by modifying the potential in $\mathbb{H}$, rather than by re-meshing or re-solving a PDE in Ω.

6 Conclusions and Future Directions

We have demonstrated that the Schwarz–Christoffel transformation provides a clean, exact framework for ideal fluid flow in an irregular polygonal domain. The forward-map approach evaluates analytically tractable potentials in $\mathbb{H}$ and maps the results to Ω, yielding artifact-free streamlines that satisfy the no-penetration condition exactly. Four physically motivated complex potentials were derived and visualised; each extends the previous by an additive term that preserves the boundary condition via the method of images.

The principal mathematical content is the interplay between conformal-mapping theory (Riemann Mapping Theorem, SC integral formula, interior angle structure, Möbius normalisation) and ideal-flow theory (complex potential, circle theorem, point vortices, method of images). The SC formula (12) encodes the polygon geometry directly through the exponents $\alpha_k - 1$: each vertex angle appears as a branch-point strength of the integrand. The Cauchy–Riemann equations, through the conformality of f, guarantee that every solution of Laplace's equation in $\mathbb{H}$ pulls back to a solution in Ω; this is the fundamental reason why conformal mapping works for fluid flow.

Numerically, the softmax reparameterisation of the pre-vertex problem transforms a constrained optimisation into an unconstrained one, enabling Levenberg–Marquardt to converge reliably. The forward-map approach avoids the per-point Newton iterations of the inverse map at the cost of a scattered-data interpolation step.

Future directions. Several extensions are natural.

- **Exact doubly-connected SC map.** The Milne-Thomson approximation (34) has error $O((a/\operatorname{Im}\zeta_0)^2)$. The achieved ratio $\operatorname{Im}(\zeta_0)/a \approx 5.36$ gives $\sim 3.5\%$ error. An exact doubly-connected SC map (Driscoll–Trefethen, Chapter 5) would give a rigorous treatment of the doubly-connected domain.
- **Other city shapes.** The pipeline is city-agnostic. Manhattan (near-rectangular) and Los Angeles (highly non-convex) would illustrate how the SC exponents encode shape complexity and crowding.
- **Multiple obstacles.** The city may contain several distinct obstacles. The Schottky-double conformal map or an iterative circle-theorem application could handle multiply-connected domains of arbitrary genus.
- **Viscous correction.** The Stokes stream function satisfies a biharmonic equation $\nabla^4\psi = 0$. Comparing ideal and Stokes streamlines would quantify the effect of viscosity near the boundary.
- **Time-dependent flow.** Adding a time-dependent circulation $\Gamma(t)$ around the domain or obstacle would model unsteady effects such as gust events, which are common in Boulder's complex terrain.

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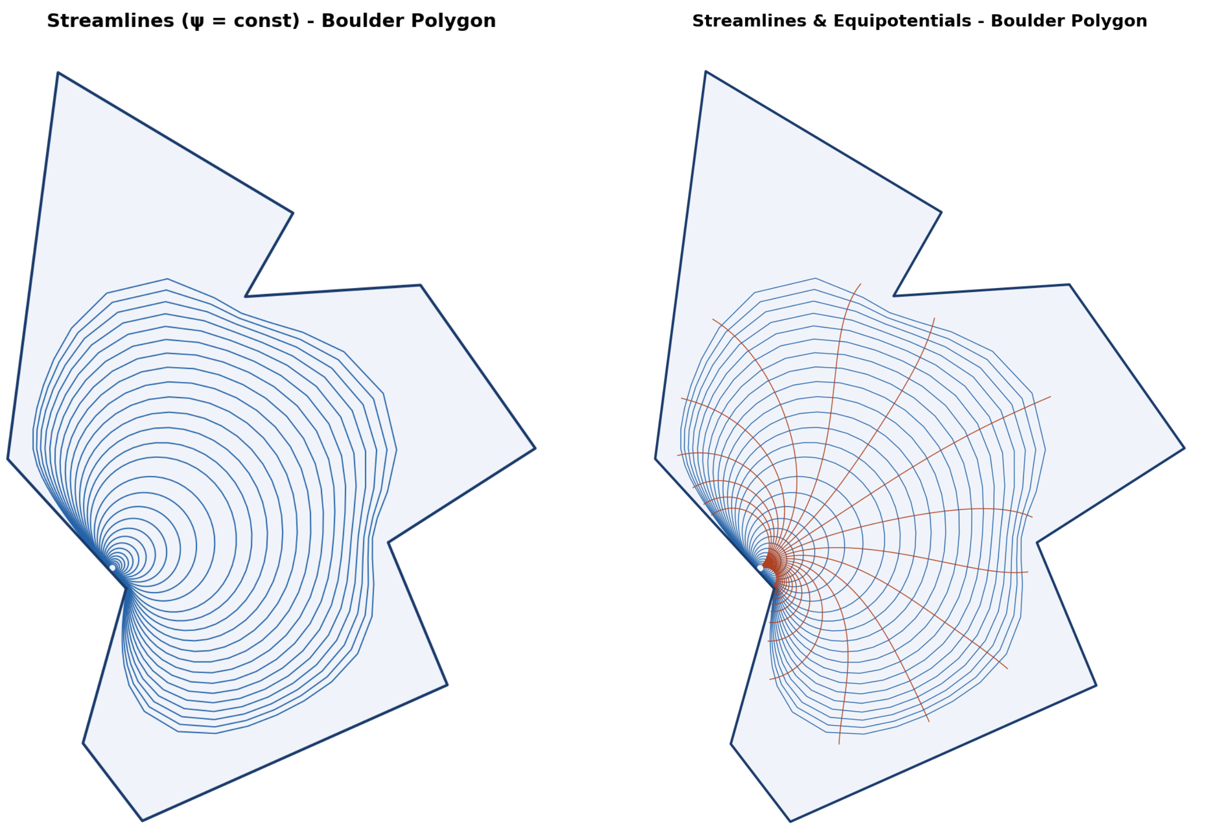
A Supplementary Figures

Per-model results referenced in Section 5 are collected here, followed by supplementary comparison panels. Detailed interpretation is given in the main body; captions below are brief labels.



(a) Streamlines $\psi = \text{const}$.

(b) Conformal grid: streamlines + equipotentials.

Figure 3: Uniform flow $W = U\zeta$ in Ω. See Section 5.2.

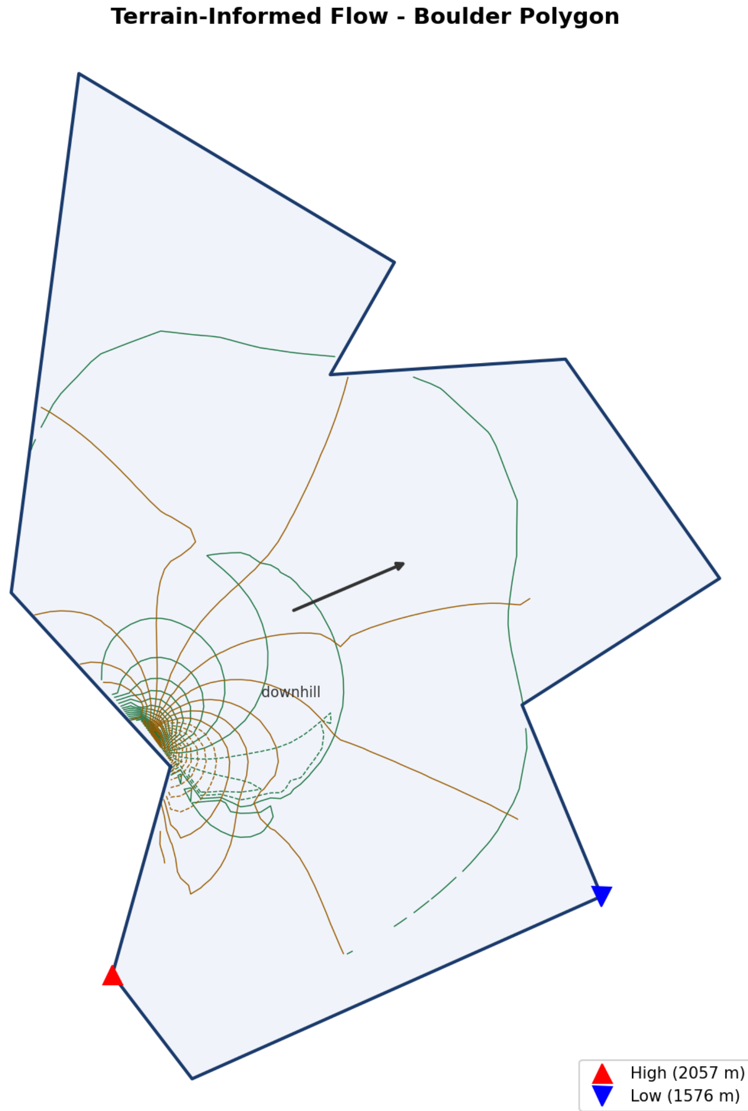


Figure 4: Terrain-corrected potential (43): streamlines (green), equipotentials (amber), high-elevation sources (red triangles), low-elevation sinks (blue triangles).

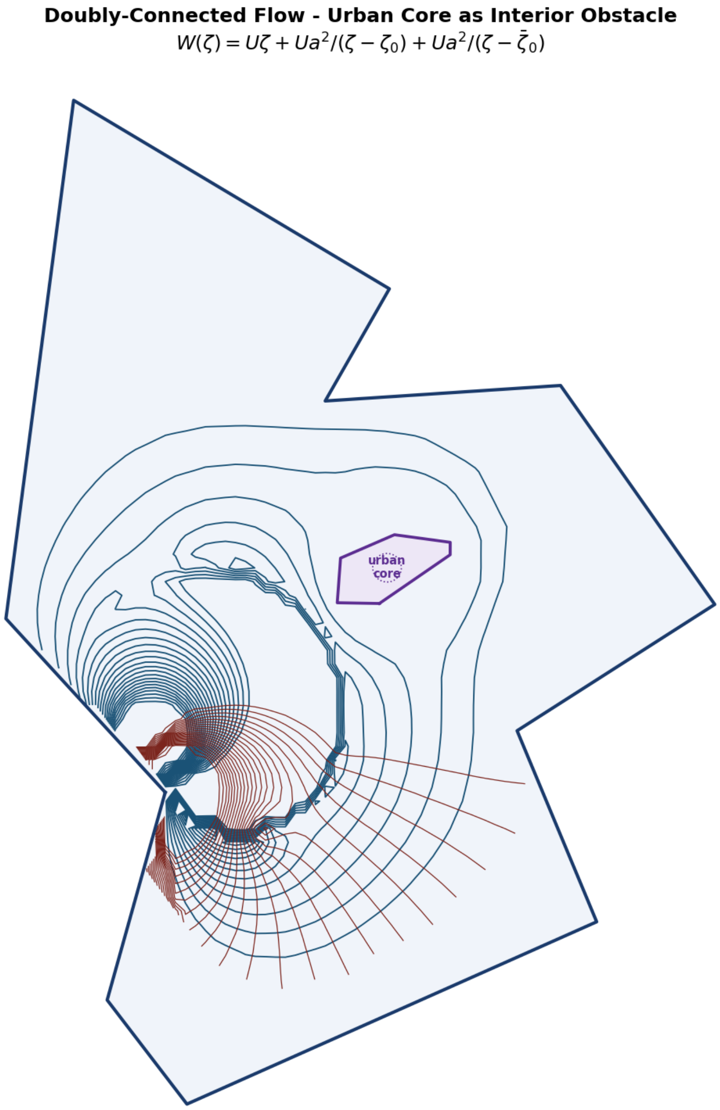


Figure 5: Urban-obstacle potential (34). Downtown core shown in purple. Separation ratio $\mathrm{Im}(\zeta_0)/a \approx 5.36$.

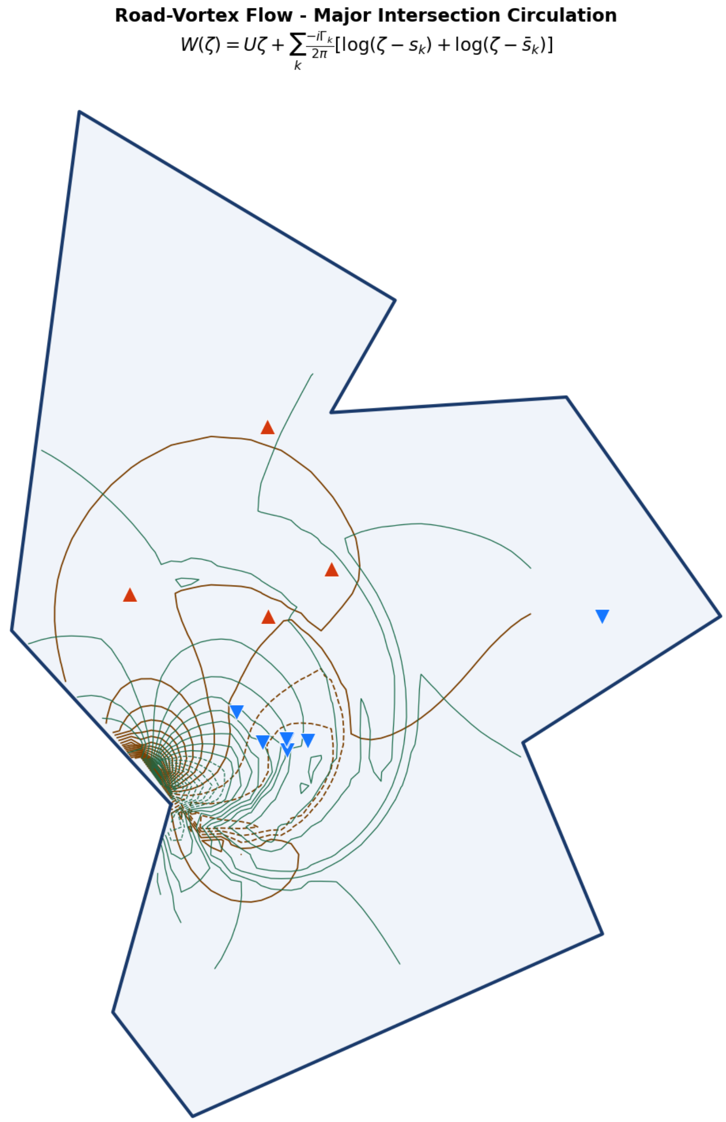


Figure 6: Road-vortex potential (44). Orange triangles: $\Gamma > 0$ (CCW, north of centroid). Blue triangles: $\Gamma < 0$ (CW, south).

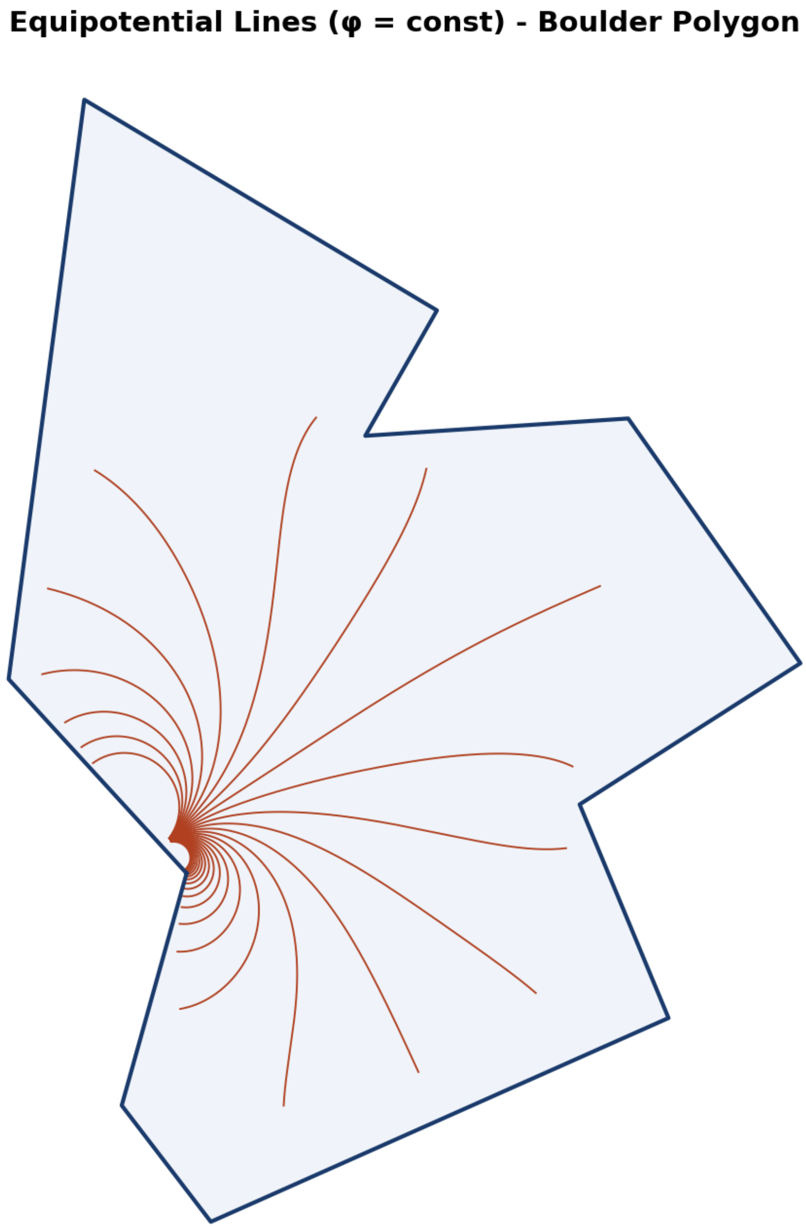


Figure 7: Equipotentials $\phi = \text{const}$ for uniform flow, shown alone.

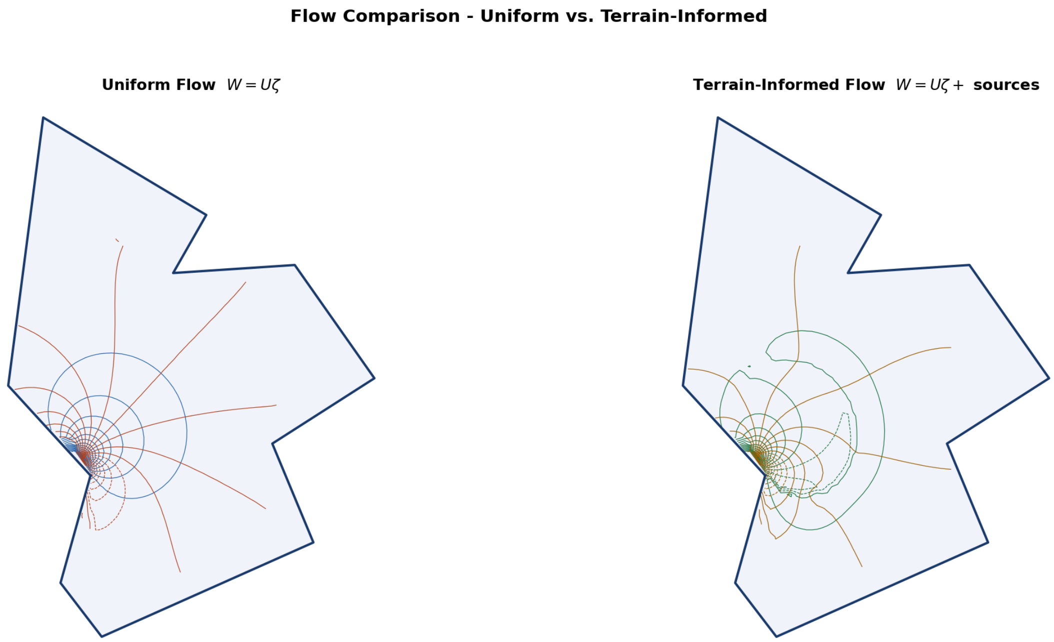


Figure 8: Uniform flow (left) vs. terrain-corrected flow (right).

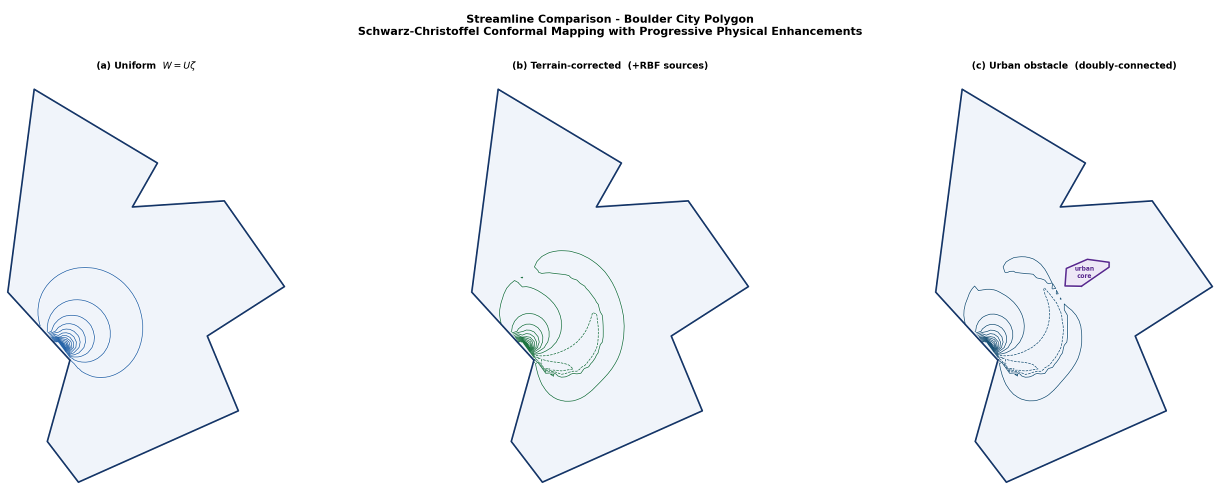


Figure 9: Three-way comparison: uniform, terrain-corrected, urban-obstacle.

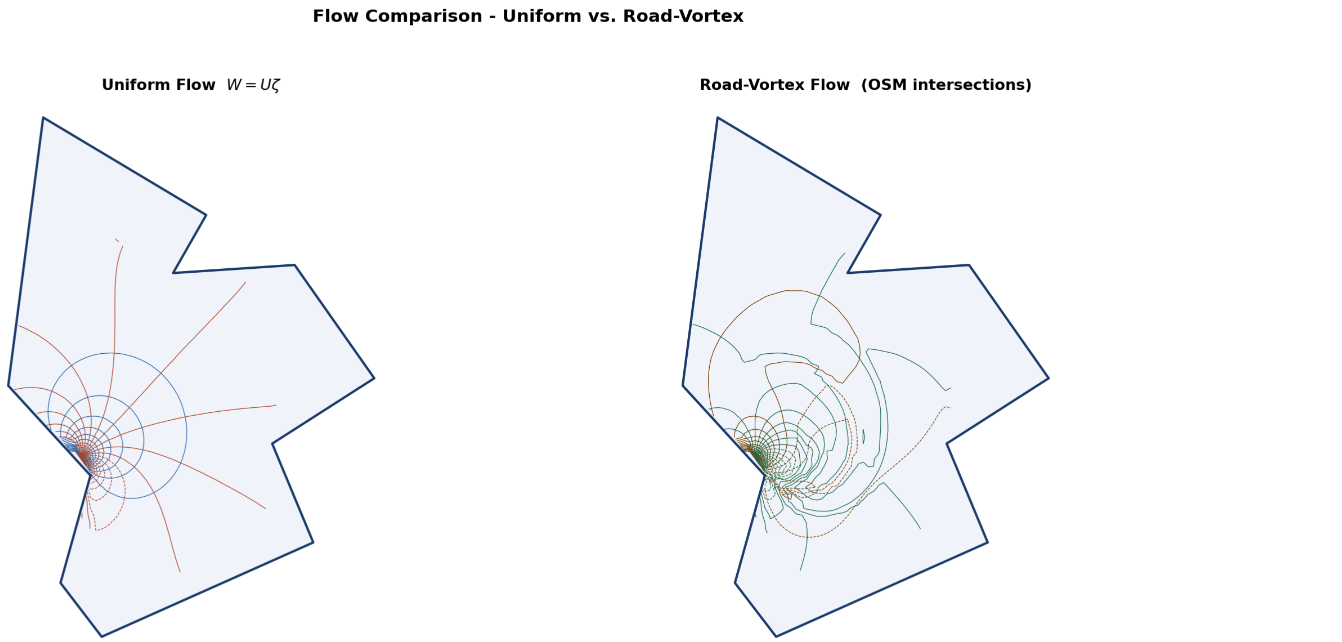


Figure 10: Uniform flow (left) vs. road-vortex flow (right).

B Code and Reproducibility

All source code is publicly available at https://github.com/McDonaldCrispyThigh/Ideal-Flow-via-Schwarz-Ch
The full pipeline is reproduced by:

```
pip install -r requirements.txt
python main.py --shapefile data/raw/tl_2025_08_place \
               --terrain --urban --roads --grid 80
```

Key modules: `src/polygon.py` (boundary preprocessing), `src/angles.py` (interior angle computation), `src/sc_solver.py` (SC parameter problem, Gauss–Legendre quadrature, forward map), `src/terrain.py` (USGS 3DEP API, RBF surface, per-vertex sources), `src/urban.py` (OSM obstacle, Welzl minimum enclosing circle), `src/roads.py` (OSM road intersections, SC inverse map, vortex placement), `src/sc_solver_dc.py` (combined potentials), and `src/visualization.py` (publication-quality figures).
Software: Python 3.14, NumPy 2.2, SciPy 1.15, Shapely 2.0, GeoPandas 1.0, OSMnx 2.0 [Boeing, 2017], Matplotlib 3.10, pyproj 3.7, tqdm 4.67.